that location can be determined. (The position of a particle and the rate of change of its position—its velocity—play analogous roles in quantum mechanics to the value of a field and the rate of change of the field value, at a given location in space.)

I like to summarize the uncertainty principle by saying, roughly speaking, that quantum mechanics makes things jittery and turbulent. If the velocity of a particle can't be delineated with total precision, we also can't delineate where the particle will be located even a fraction of a second later, since velocity *now* determines position *then.* In a sense, the particle is free to take on this or that velocity, or more precisely, to assume a mixture of many different velocities, and hence it will jitter frantically, haphazardly going this way and that. For fields, the situation is similar. If a field's rate of change can't be delineated with total precision, then we also can't delineate what the value of the field will be, at any location, even a moment later. In a sense, the field will undulate up or down at this or that speed, or, more precisely, it will assume a strange mixture of many different rates of change, and hence its value will undergo a frenzied, fuzzy, random jitter.

In daily life we aren't directly aware of the jitters, either for particles or fields, because they take place on subatomic scales. But that's where inflation makes a big impact. The sudden burst of inflationary expansion stretched space by such an enormous factor that what initially inhabited the microscopic was drawn out to the macroscopic. As a key example, pioneers[1] of inflationary cosmology realized that random differences between the quantum jitters in one spatial location and another would have generated slight inhomogeneities in the microscopic realm; because of the indiscriminate quantum agitation, the amount of energy in one location would have been a bit different from what it was in another. Then, through the subsequent inflationary swelling of space, these tiny variations would have been stretched to scales far larger than the quantum domain, yielding a small amount of lumpiness, much as tiny wiggles drawn on a balloon with a Magic Marker are stretched clear across the balloon's surface when you blow it up. This, physicists believe, is the origin of the lumpiness that the staunch believer in the standard big bang model simply declares, without justification, to be "how it was back then." Through the enormous stretching of inevitable quantum fluctuations, inflationary cosmology provides an explanation: inflationary